Advanced Summary Sheet: Game Theory

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Strategic Tips & Dangers (Read First!)

- **Game Type:** Immediately identify if the game is **Zero-Sum** ($\sum u_i = 0$) or not. Theorems (Sion vs Glicksberg) and methods (Minimax vs Nash) differ.
- **Existence:** Check compactness and continuity assumptions. Glicksberg requires quasi-concavity of players. Sion has asymmetric conditions (quasi-concave/quasi-convex).
- **Optimization:** Using the First Order Condition (FOC) is not enough! You must **always** verify the Second Order Condition (SOC) or argue the concavity of the utility (payoff) function for the player considered.
- **Pure vs Mixed:** Do not confuse pure and mixed equilibrium. A mixed equilibrium without a unique pure support implies indifference between the strategies of the support.
- **Non-existence:** There are games without value (e.g., Sion-Wolfe) or without Nash equilibrium (if discontinuity or non-quasi-concavity).

Methodology: Detecting Non-Existence

Suspect non-existence of equilibrium or value if:

- **Discontinuity:** The payoff function has a jump (e.g., auctions, war of attrition, Sion-Wolfe).
- **Non-compactness:** The space is open (e.g., $]0, 1[$) or unbounded ($[0, \infty[$) and payoffs grow to infinity.
- **Non-Concavity:** The utility (payoff) function u_i is not quasi-concave (multiple bumps), making FOCs insufficient.

1 Normal Form Games

A game is a triplet $G = (N, (X_i)_{i \in N}, (u_i)_{i \in N})$.

- **Dominance:** An action x_i^* is strictly dominant if:

$$\forall x_{-i} \in X_{-i}, \forall x_i \in X_i \setminus \{x_i^*\}, \quad u_i(x_i^*, x_{-i}) > u_i(x_i, x_{-i})$$

- **IESDS (Iterated Elimination of Strictly Dominated Strategies)**: Sequentially eliminate strictly dominated strategies. If the game is finite and solvable by dominance, the order of elimination does not matter.
- **Rationalizable Strategies:** These are the strategies that survive the iterated elimination of strategies that are never best responses.
- **Crucial Link:** In a 2-player game, the set of rationalizable strategies coincides exactly with that resulting from IESDS.
- **Dominant Strategy Equilibrium (DSE):** A profile s^* is a DSE if for each player i, s_i^* is a best response regardless of the behavior of others ($u_i(s_i^*, s_{-i}) \geq u_i(s_i, s_{-i})$ for all s_{-i}).
- *Exam Note:* Every DSE is a Nash equilibrium, but the converse is false.

Exam Rigor: IESDS Wording

To justify an elimination at step k: *"In the reduced game $G(k)$, for player i, action A is strictly dominated by action B because for any opposing profile $x_{-i} \in X_{-i}(k)$, we have $u_i(A, x_{-i}) < u_i(B, x_{-i})$."*

- **Nash Equilibrium (NE):** A profile $\bar{x} = (\bar{x}_1, \dots, \bar{x}_n)$ is an NE if no player has an incentive to unilaterally deviate:
$$\forall i \in N, \forall x_i \in X_i, \quad u_i(\bar{x}_i, \bar{x}_{-i}) \geq u_i(x_i, \bar{x}_{-i})$$
In other words, $\bar{x}_i$ maximizes $x_i \mapsto u_i(x_i, \bar{x}_{-i})$.

Distinction: Pure vs Mixed Equilibrium

- **Pure:** Each player chooses ONE single action. You are looking for $x_i^* \in X_i$.
- **Mixed:** Each player chooses a PROBABILITY DISTRIBUTION σ_i. You are looking for $p \in [0, 1]$.
- **Hint:** If asked to "show that there is no pure equilibrium", look for a best response cycle $A \to B \to C \to A$. The equilibrium will then be mixed.

2 The 3 Pillars of Existence Proof (Exercise Structure)

1. Weierstrass Theorem (Extremum)

Used to justify that the set of best responses BR_i is not empty.

- **Assumptions:**
  - The set of strategies X_i is **compact** (closed and bounded).
  - The payoff function u_i is **continuous** with respect to their own strategy.
- **Result:**
  - The maximum of the function is reached on the set.
  - The set of maximizing arguments (the argmax) is therefore **non-empty**.

2. Kakutani Fixed Point Theorem

Fundamental tool to prove the existence of an equilibrium. Applied to the correspondence $B : X \rightrightarrows X$.

- **Assumptions:**
  - **On the space** X: Must be compact, convex, and non-empty in $\mathbb{R}^n$.
  - **On the correspondence** B:
    1. **Non-empty** for all x (ensured by Weierstrass).
    2. **Convex values** (ensured by quasi-concavity of utility/payoff).
    3. **Closed graph** (ensured by continuity of utility/payoff / Berge's Thm).
- **Result:**
  - There exists a fixed point $x^* \in X$ such that $x^* \in B(x^*)$.
  - This fixed point is a **Nash Equilibrium**.

3 Continuous Games and Existence Theorems

3.1 Practical Resolution Method

To find an NE in a game with continuous strategies (e.g., Cournot, Bertrand): 1. **FOC:** Calculate $\frac{\partial u_i}{\partial x_i}(x_i, x_{-i}) = 0$. 2. **Best Response:** Isolate x_i to express $BR_i(x_{-i})$. 3. **System:** Solve the system $x_i = BR_i(x_{-i})$ for all

i. 4. **Verification (Crucial):** Prove that u_i is concave in x_i (e.g., $\frac{\partial^2 u_i}{\partial x_i^2} < 0$), ensuring the FOC gives a global maximum.

FOC/SOC Rigor

To justify that a critical point is a global maximum, it is imperative to verify the Second Order Condition (SOC) or concavity. *Example:* "The function u_i is concave with respect to x_i because $\frac{\partial^2 u_i}{\partial x_i^2} < 0$, so the first order condition is necessary and sufficient for a global maximum".

3. Glicksberg's Theorem (Game Synthesis)

"Game Theory" version guaranteeing the existence of a pure strategy NE.

- **Assumptions:**
  - **Convexity and Compactness:** Action sets X_i are non-empty, convex, and compact.
  - **Continuity:** Payoff functions u_i are continuous (in (x_i, x_{-i})).
  - **Quasi-concavity:** Each u_i is quasi-concave with respect to its own action x_i.
- **Result:**
  - The game admits at least one Nash equilibrium.

Why is quasi-concavity crucial?

It is indispensable because it guarantees that the set of best responses $B(y)$ is **convex**. This fulfills one of the *sine qua non* conditions of Kakutani's theorem for the existence of an equilibrium. *(Method: To prove convexity of $B(y)$, take $x_1, x_2 \in B(y)$ and show that $\lambda x_1 + (1-\lambda)x_2$ is also in $B(y)$ via quasi-concavity).*

3.2 Unbounded Case (Coercivity)

If $X_i = [0, +\infty[$, compactness is missing. We use a coercivity argument: If $\lim_{x_i \to +\infty} u_i(x_i, x_{-i}) = -\infty$ (uniformly or for relevant candidates), we can restrict the study to a sufficiently large compact $[0, K]$ without losing potential equilibria.

4 Extensive Form Games and SPE

- **Strategy:** Complete plan of actions defined for *every* information set (node), even those that will never be reached under that same strategy.
- **Kuhn-Zermelo Theorem:** Every finite game with perfect information possesses a Nash equilibrium in pure strategies, which can be found by backward induction (SPE).
- **Backward Induction:** Solve the game starting from the final nodes.
- **SPE (Subgame Perfect Equilibrium):** A strategy profile is an SPE if it induces a Nash equilibrium in every subgame (reached or not).
- **Non-Credible Threat Critique:** A simple NE can rely on "incredible" threats (e.g., saying "I will blow up the game if you enter", whereas if the other enters, blowing up gives a worse payoff). The SPE eliminates these equilibria by enforcing sequential rationality.

4.1 Counter-examples (Failure of SPE Existence)

- **Infinite number of actions:** If the action space is not compact or payoffs are not continuous (e.g., game where you want to pick the highest real number, there is no maximum).
- **Infinite horizon with increasing payoffs:** If $\alpha_{n+1} > \alpha_n$, each player always wants to wait for the next turn to win more, so no one ever stops, but "never stopping" gives 0, which is worse than an immediate stop $\implies$ Contradiction, no SPE.

4.2 SPE in Infinite Horizon

Consider a game where two players alternate. Stopping at round n gives (α_n, β_n). Continuing indefinitely gives $(0, 0)$.

- **Case 1: Strictly increasing payoffs** $(\alpha_{n+1} > \alpha_n)$. **Theorem: No SPE exists.** *Proof by contradiction:* Suppose stopping at round N. The active player at N would prefer to "pass" to let the other stop (or stop themselves) later to win more $(\alpha_{N+2} > \alpha_N)$. If no one ever stops, payoff 0. But stopping at 1 gives $\alpha_1 > 0$, so "never stopping" is not optimal. Contradiction.
- **Case 2: Strictly decreasing payoffs** $(\alpha_{n+1} < \alpha_n)$. **Theorem: There exists an SPE (immediate stop).** Since future payoffs are always less than the present payoff, the player to move has an incentive to secure the current payoff.

5 Zero-Sum Games $(u_1 + u_2 = 0)$

Tip: Spotting a Zero-Sum Game

Check if $u_1(x, y) + u_2(x, y) = C$ (constant, often 0). If yes, it is a strictly competitive game: what one gains is lost by the other. **Consequence:** We can use Sion's theorem and look for the Value (Maximin = Minimax). If not zero-sum, we look for a classic Nash.

Denote $u(x, y) = u_1(x, y)$. Player 1 (P1) maximizes u, P2 minimizes u (since maximizing $-u$).

- **Maximin (P1 Security):** $\underline{v} = \sup_{x \in X} \inf_{y \in Y} u(x, y)$. What P1 can guarantee in the worst case.
- **Minimax (P2 Security):** $\bar{v} = \inf_{y \in Y} \sup_{x \in X} u(x, y)$. What P2 can limit to at best if P1 plays perfectly.
- **Value of the Game:** The game has a value if $\underline{v} = \bar{v}$. In this case, there exists (x^*, y^*) such that $u(x^*, y^*) = v$. These strategies are optimal ("saddle point").

Equivalence Theorem: A game possesses a value v if and only if it possesses a Nash equilibrium (x^*, y^*). In this case, $v = u(x^*, y^*)$ and the equilibrium strategies are the optimal strategies (maximin/minimax).

Sion's Theorem (Existence of a Value)

Let $G = (X, Y, u)$ be a zero-sum game where:

- X is the set of strategies for Player 1 (maximizes u).
- Y is the set of strategies for Player 2 (minimizes u).

The game possesses a **Value** $(\sup \inf = \inf \sup)$ if:

1. X and Y are **convex** sets in topological vector spaces.
2. **One of the two** spaces is **compact**.
3. $x \mapsto u(x, y)$ is **quasi-concave** and **u.s.c.** (upper semi-continuous) on X.
4. $y \mapsto u(x, y)$ is **quasi-convex** and **l.s.c.** (lower semi-continuous) on Y.

Key Steps of Sion's Proof:

1. *Contradiction:* Assume no value, so $\underline{v} < \bar{v}$. Insert a real c between the two.
2. *Covering:* Use semi-continuity properties to define open sets. Thanks to compactness (say of Y), extract a finite subcover.
3. **Intersection Lemma (Generalised Knaster-Kuratowski-Mazurkiewicz):** If a family of compact convex sets has the property that any intersection of k elements contains the convex hull of... (simplified version: non-empty intersection). Use this lemma on level sets to show that players cannot "escape" value c, leading to a contradiction.

Intersection Lemma (Key to Sion)

Let $A_1, \ldots, A_n$ be **convex compact** subsets. If $\cup_{i=1}^n A_i$ is convex and if all intersections of $n-1$ sets are non-empty ($\cap_{i \neq j} A_i \neq \emptyset$), then the total intersection is non-empty ($\cap_{i=1}^n A_i \neq \emptyset$). *Exercise Note:* Specify that sets are "convex and compact in a Euclidean space".

6 Mixed Strategies

If no pure equilibrium, look for probability measures σ_i.

- **Theorem:** In a finite game, there always exists a mixed strategy NE (Nash 1950).
- **Support:** Set of actions played with positive probability.
- **Indifference Principle:** If a player plays a mixed strategy at equilibrium, they must be **indifferent** between all pure actions in their support. All these actions must yield the same expected payoff (equal to the game payoff for this player).

2x2 Calculation Method: Let P1 play Top (p) / Bottom ($1-p$) and P2 play Left (q) / Right ($1-q$). To find q^* (P2 mix), write P1's indifference:

$$E[u_1(T, \sigma_2)] = E[u_1(B, \sigma_2)]$$

$$q \cdot u_1(T, L) + (1-q) \cdot u_1(T, R) = q \cdot u_1(B, L) + (1-q) \cdot u_1(B, R)$$

Solve for q. Same for p with P2's payoffs. **Warning:** P1's probability p makes P2 indifferent (depends on P2's payoffs).

7 Classic Counter-Examples

- **Sion-Wolfe Game (Technical Detail):** Typical example of a zero-sum game without value (and without mixed equilibrium). *Reason for failure:* The utility (payoff) function is discontinuous (often a step function along the diagonal), preventing the application of Sion's theorem because semi-continuity conditions (usc/lsc) are violated.
- **Centipede Game:** Finite extensive form game. *Paradox:* Backward induction (SPE) predicts the first player stops immediately (payoffs $(1,1)$ perhaps), whereas continuing would lead to $(100,100)$. Conflict between sequential individual rationality and Pareto efficiency.